Module 3 Lab 1: PII Redaction Pipeline with Amazon Comprehend

20 points. Estimated time 90 to 120 minutes.

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Date: 11/5/26

How to use this file. Type your answers straight into the shaded boxes. Do not delete the questions. Anything marked as written in advance has to be filled in before you run that step, not afterwards. When you are finished, save this file and upload it with the rest of your submission.

1. Your unique values

These are what make the submission yours. They come from your own account and your own run, so no two students have the same ones.

Your Region: Us-east-2

Your S3 bucket name: cs79f-m3.1-support-tickets

The lowest confidence score you saw on any detection:
0.99

Your chosen confidence cutoff and the one sentence reason for it:
0.92 in case there is some uncertainty

The line number where something personal was still visible, if you found one:
NA

2. Predictions, written in advance

Write each prediction BEFORE you run that step. A wrong prediction reasoned honestly earns full credit. One edited afterwards to match the result earns none.

Step 1

Before you choose Analyze, predict this: will the Entities tab list MORE items, FEWER items, or the SAME items as the PII tab for this ticket?

The Same

What actually happened

They had the same items

Why it was different, or why I was right

They both collect person/name and contact info, and no other entities were present

3. Break it on purpose

Working code shows you what happens. Broken code shows you why. Do each of these deliberately, paste the real error text, and undo the change before moving on.

Break 1. Create a third Redactions job, but under PII entity types select only EMAIL. Run it and open the output.

Paste the exact error text here, not a description of it

What is now readable that should not be. Name the types that came through and say what a support agent could do with them.

Names, phone numbers, credit card info are available. A support agent could harass the person or impersonate them, or sell their information.

Break 2. In Real time analysis, paste a ticket but rewrite the person's name in all lowercase with no punctuation, for example sam reyes. Analyze it again.

Paste the exact error text here, not a description of it

Whether NAME was still found, and its confidence score before and after. Write both numbers down.

The name was found, confidence was 0.99+ both times

Break 3. Create a job whose Output data location points at a bucket that is not yours, for example a name you invent.

Paste the exact error text here, not a description of it

Failure reason
OUTPUT_BUCKET_NOT_FOUND: The provided output S3 bucket does not exist.

The exact status the job list shows and the exact error text on the job detail page. Do not guess at it, read it.

It says failed. The error text is: **Failure reason**
OUTPUT_BUCKET_NOT_FOUND: The provided output S3 bucket does not exist.

4. Results

The numbers and outputs your own run produced.

Your Region and your S3 bucket name

Us-east-2
cs79f-m3.1-support-tickets

Your detection list from Steps 2 and 3, with confidence scores

Sam Reyes	Person	0.99+
sam.reyes@example.com	Other	0.99+
310-555-5787	Other	0.99+
Victor Okafor	Person	0.99+
victor.okafor@example.com	Other	0.99+
310-555-8858	Other	0.99+
dana.kim@example.com	Other	0.99+
Dana Kim	Person	0.99+
310-555-9558	Other	0.99+
Tom Moreno	Person	0.99+

Your chosen confidence cutoff and its one sentence reason

0.92 in case there is some uncertainty in categorization

Your six count lines from Step 6

{"Entities": [{"BeginOffset": 12, "EndOffset": 21, "Type": "NAME", "Score": 0.9999947548200224}, {"BeginOffset": 69, "EndOffset": 90, "Type": "EMAIL", "Score": 0.9999995231630692}, {"BeginOffset": 94, "EndOffset": 106, "Type": "PHONE", "Score": 0.999994397195003}], "File": "tickets.txt", "Line": 0}
{"Entities": [{"BeginOffset": 12, "EndOffset": 25, "Type": "NAME", "Score": 0.9999364664510629}, {"BeginOffset": 73, "EndOffset": 98, "Type": "EMAIL", "Score": 0.9999990463265931}, {"BeginOffset": 102, "EndOffset": 114, "Type": "PHONE", "Score": 0.9999957084838798}], "File": "tickets.txt", "Line": 1}
{"Entities": [{"BeginOffset": 35, "EndOffset": 55, "Type": "EMAIL", "Score": 0.9999988079084972}, {"BeginOffset": 68, "EndOffset": 76, "Type": "NAME", "Score": 0.9999954104637981}, {"BeginOffset": 94, "EndOffset": 106, "Type": "PHONE", "Score": 0.9999933243243502}], "File": "tickets.txt", "Line": 2}
{"Entities": [{"BeginOffset": 12, "EndOffset": 22, "Type": "NAME", "Score": 0.9999958872963806}, {"BeginOffset": 70, "EndOffset": 92, "Type": "EMAIL", "Score": 0.9999995231630692}, {"BeginOffset": 96, "EndOffset": 108, "Type": "PHONE", "Score": 0.9999918938339075}], "File": "tickets.txt", "Line": 3}
{"Entities": [{"BeginOffset": 10, "EndOffset": 22, "Type": "NAME", "Score": 0.999997615820007}, {"BeginOffset": 61, "EndOffset": 76, "Type": "ADDRESS", "Score": 0.9999920925495949}, {"BeginOffset": 82, "EndOffset": 94, "Type": "PHONE", "Score": 0.9999958276922101}], "File": "tickets.txt", "Line": 4}
{"Entities": [], "File": "tickets.txt", "Line": 5}

5. Reflection

Your own words. No AI assistance on this part. Graders check these answers against the rest of your submission.

Question 1

Paste the redacted text of the one line where something personal was still visible, or if everything was redacted, the line with the lowest confidence score you saw. Explain what about that specific text made it hard for the service, and name one concrete change you would make to catch it.

The lowest confidence score I saw was 0.99+. The model is very effective on this data.

Question 2

Defend your confidence cutoff from Step 3 as though a healthcare compliance officer is reading it. Name the harm of a missed detection and the harm of a false one, say which you weighted more heavily, and quote your own lowest confidence score as part of the argument.

The harm of a FP is a possible expensive lawsuit, while the harm of a FN is just inconvenience to the support staff and possibly some extra man hours.

6. AI use disclosure

Required by the course AI Use Policy.

Tool used (write none if you used no AI):

If you used one: what you asked it, what you kept, and how you checked it

none

7. Before you submit

- ☐ I checked twelve lines before I opened the answer key column
- ☐ My tickets.txt had sixty lines and no header
- ☐ My cutoff has a reason that names both harms
- ☐ I ran this entirely in the console and wrote no code
- ☐ Every prediction was written before I ran that step
- ☐ I recorded the actual result for every prediction, including the wrong ones
- ☐ I pasted the real error text for every break it item
- ☐ Work pushed to my GitHub portfolio repository and the URL is in here

- ☐ Reflection written by me with no AI assistance
- ☐ Files uploaded individually, no archive